

**CS433: Internet of Things**  
**IoT Final Project Report**

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**Smart Irrigation System**

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*Smart City — Egypt's New Administrative Capital*

**Faculty of Computers and Artificial Intelligence**  
Benha University — 2026

# 1.Introduction

Egypt's new administrative capital is designed as a smart city that integrates advanced IoT infrastructure to serve millions of citizens. One of its most critical challenges is efficient water management. Traditional irrigation in public parks, green spaces, and urban gardens wastes significant amounts of water through fixed schedules that ignore actual soil conditions and weather. This not only depletes scarce water resources but also increases operational costs. The Smart Irrigation System addresses this challenge by automating irrigation decisions based on real-time environmental sensor data, ensuring that water is applied only when and where it is needed.

This project presents an IoT-based Smart Irrigation System that monitors soil moisture, temperature, and humidity in real time using sensors connected to an ESP microcontroller. The system uses MQTT communication to transfer data to a web dashboard where users can monitor sensor readings and control the water pump remotely.

## 2. Problem Statement

Traditional irrigation systems suffer from several problems:

- Excessive water consumption
- Manual monitoring and operation
- Lack of real-time environmental monitoring
- Difficulty controlling irrigation remotely

These issues lead to inefficient water usage and reduced agricultural productivity.

## 3. Objectives

- Monitor soil moisture, air temperature, and humidity in real time using IoT sensors.
- Automatically control a water pump based on configurable soil moisture thresholds (Auto Mode).
- Allow operators to override automation and control the pump manually (Manual Mode).
- Transmit sensor data from edge devices to a web dashboard over MQTT.
- Store and visualize sensor data in real time.
- Provide a clear, user-friendly web interface accessible from any device.
- Reduce water consumption and operational costs in urban green spaces.

## 4. Scope

The project covers the design, implementation, and testing of a complete end-to-end smart irrigation system. It includes the device/sensor layer, communication layer using MQTT over Wi-Fi, cloud data storage, and a web-based dashboard. The system targets outdoor green areas of the new administrative capital, though its architecture is scalable to multiple zones.

The following are explicitly within scope:

- Smart irrigation automation
- Real-time environmental monitoring
- Remote web-based control
- MQTT-based IoT communication
- Cloud integration using Firebase

The system is designed for small farms, gardens, and smart agriculture applications.

## 5. Proposed Solution

The proposed system uses sensors connected to an ESP microcontroller to collect environmental data. The ESP sends the readings using MQTT protocol to a cloud-connected web dashboard.

The dashboard displays:

- Soil moisture percentage
- Temperature
- Humidity
- Pump status
- System mode (Auto/Manual)

When the soil moisture becomes lower than a predefined threshold, the system automatically activates the water pump using a relay module.

Users can also manually control the pump from the dashboard.

## 6. Inputs and Outputs

| Type   | Component            | Description                                                                      |
|--------|----------------------|----------------------------------------------------------------------------------|
| INPUT  | Soil Moisture Sensor | Measures soil water level                                                        |
| INPUT  | Temperature Sensor   | Measures temperature                                                             |
| INPUT  | Humidity Sensor      | Measures air humidity                                                            |
| OUTPUT | Relay Module         | Controls pump switching                                                          |
| OUTPUT | MQTT Sensor Data     | JSON payload published every 5 s containing moisture, temperature, and humidity. |
| OUTPUT | Water Pump           | Irrigates plants                                                                 |
| OUTPUT | Web Dashboard        | Displays sensor data and controls                                                |

## 7. Sensors and Actuators

| Component            | Type     | Function                          |
|----------------------|----------|-----------------------------------|
| Soil Moisture Sensor | Sensor   | Detects soil moisture level       |
| DHT11                | Sensor   | Measures temperature and humidity |
| 5V Relay Module      | Actuator | Switches the water pump ON/OFF    |
| Water Pump           | Actuator | Supplies water to plants          |

|         |                 |                                            |
|---------|-----------------|--------------------------------------------|
| ESP8266 | Microcontroller | Processes data and communicates using MQTT |
|---------|-----------------|--------------------------------------------|

## 8. Communication Protocol

The system uses MQTT protocol for lightweight real-time communication between devices and the web dashboard.

MQTT Broker Used:

- HiveMQ Public Broker

Communication Topics:

- Soil moisture topic
- Temperature topic
- Humidity topic
- Pump control topic
- Mode control topic

## 9. System Architecture

The system is organized into three layers:

| Layer               | Components                                                                            |
|---------------------|---------------------------------------------------------------------------------------|
| Device Layer        | Moisture Sensor - DHT11 - 5V Relay Module - Water Pump                                |
| Communication Layer | ESP8266 - MQTT protocol - HiveMQ broker                                               |
| Application Layer   | Web Dashboard (HTML/CSS/JS) - Real-time charts and controls - Firebase cloud database |

## 10. Block Diagram

**Sensors → ESP8266/ESP32 → MQTT Broker → Web Dashboard**



**Firebase Storage**



**Relay Module → Pump**

## 11. User Interface Features

The web dashboard provides:

- Real-time sensor monitoring
- Automatic/manual mode switching
- Pump ON/OFF control
- Live charts
- Cloud synchronization
- Responsive user interface

## 12. Test Strategy

The system was tested using different scenarios to verify correct operation.

| Test Case                | Expected Result               | Status |
|--------------------------|-------------------------------|--------|
| Dry soil condition       | Pump turns ON automatically   | Passed |
| Wet soil condition       | Pump remains OFF              | Passed |
| Manual pump ON           | Pump activates from dashboard | Passed |
| Manual pump OFF          | Pump stops from dashboard     | Passed |
| MQTT connection          | Data transferred successfully | Passed |
| Firebase synchronization | Dashboard updates correctly   | Passed |
| Sensor reading update    | Real-time values displayed    | Passed |

## 13. Benefits of the System

- Saves water
- Reduces manual effort
- Provides remote monitoring
- Improves irrigation efficiency
- Supports smart agriculture applications
- Real-time data analysis and visualization

## 14. Future Improvements

Possible future improvements include:

- Mobile application support
- AI-based irrigation prediction
- Weather forecast integration
- Solar-powered operation
- SMS or notification alerts

## 15. Conclusion

The Smart Irrigation System successfully demonstrates the integration of IoT technologies in agriculture. The project combines sensors, MQTT communication, cloud services, and a web dashboard to create an efficient irrigation solution. The system helps reduce water waste, automate irrigation processes, and improve monitoring capabilities.